PRABHJOT (Prabh) SINGH

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SUMMARY OF QUALIFICATIONS

AWS Certified Associate Developer with **6+ YoE** building scalable, secure and high-performance backend systems using Python, AWS, and modern cloud-native technologies. Proven track record leading the delivery of major product features and improving system performance across high-throughput, real-time media platforms utilizing Gen AI. Experienced in designing and maintaining distributed systems for ingesting and processing large volumes of structured, unstructured, and video-based data, building and deploying AI agents using FastAPI, and designing MCP servers exposing broadcast metadata. Supported by a strong background in DevOps, CI/CD pipelines, GitHub Actions, Infrastructure as Code (IaC), cloud automation and SDLC practices, while working in an Agile Scrum environment to ship production-ready solutions.

EXPERIENCE

Evertz **Jan 2023 - Present**

Software Development Engineer ***Toronto, ON***

- *Owned and modernized 4+ critical backend services* through strategic dependency management and package manager migrations, significantly improving system reliability and long-term maintainability.
- *Spearheaded the delivery* of scalable **RESTful** and WebSocket content streaming services (Python, FastAPI), optimizing data layers across Aurora PostgreSQL and DynamoDB to guarantee high-performance streaming.
- *Accelerated EC2 launch times by 30%* by engineering a GitHub Actions CI/CD pipeline featuring automated semantic versioning, CodeArtifact integration, and pre-built AMIs.
- *Designed and deployed Model Context Protocol (**MCP**)* servers to securely expose proprietary Media Asset Management (MAM) databases and live playout schedules to Large Language Models, enabling secure, contextual data querying.
- *Enabled real-time operational insights* and high-throughput system monitoring by engineering an asynchronous, event-driven FastAPI service packaged as a Python PEX on EC2 instances.
- *Engineered autonomous AI agents using FastAPI* to actively monitor broadcast routing telemetry and troubleshoot IP routing errors in real-time, reducing the need for manual operator intervention.
- *Achieved consistent sub-second SLAs* for media-fragment processing by architecting an autoscalable **FastAPI** microservice on EKS, powered by Amazon Kinesis for high-throughput event ingestion.
- *Automated media asset classification and content validation* by architecting an OpenAI driven workflow, executing real-time visual assertions on video streams to ensure broadcast compliance and operational integrity.
- *Reduced monthly cloud spend* by 35% by writing automated Python scripts to identify and terminate orphaned AWS resources, media connects, and outdated AMIs across dev/test environments.
- *Accelerated root-cause analysis* and reduced MTTR by 25–30% by implementing comprehensive distributed tracing and observability using Honeycomb and **OpenTelemetry**.
- *Conducted research* and implemented a proof of concept (PoC) for AWS Dedicated Hosts, reducing vendor license requirements by 30%, optimizing resource utilization, and cutting costs by 25%.
- *Streamlined backend service* integration by establishing a contract-first API development lifecycle (Swagger/**OpenAPI**) that seamlessly integrated into IaC workflows (CloudFormation, Terraform, CDK).
- *Delivered highly reliable production code* by establishing PyTest frameworks and driving cross-functional QA collaboration to achieve and maintain 95% end-to-end test coverage.

Deloitte USI – CIBC Bank **May 2021 - Apr 2022**

Software Development Analyst ***India***

- *Developed ETL pipelines* using **AWS Glue** with PySpark and Pandas to load and transform data from various sources into AWS S3 DataLake.
- *Architected and scaled a high-throughput bank chargeback system*, processing over 10 million daily requests with a distributed, event-driven backend. Engineered high-concurrency data pipelines to aggregate and transmit consolidated data to Mastercard and Visa, ensuring strict compliance with financial settlement protocols.
- *Optimized system responsiveness for a high-volume chargeback engine* by implementing a distributed **Redis** cache, reducing database read contention and ensuring sub-millisecond data retrieval across 10 million daily user requests.
- *Migrated database and services* to a new schema, removing backward compatibility code and reducing API load time by 20%, significantly improving performance.
- *Developed and deployed serverless* RESTful solutions using AWS Lambda and API Gateway along with AWS SAM.
- *Worked on data dimensional* modelling techniques like start schema.
- *Crafted complex SQL queries*, employing self-joins and correlated sub-queries for data manipulation and retrieval.

- *Utilized AWS Redshift* for data warehousing and contributed to Airflow workflow management.

Accenture – NBN Australia **Aug 2019 - May 2021**
Associate Software Developer ***India***

- *Architected and developed* serverless and stateless solutions on AWS.
- *Developed RESTful and Websocket* microservices using AWS Lambda, API Gateway, DynamoDB, Aurora MySQL, and Python to perform CRUD operations and manage customer data efficiently.
- *Worked on Python frameworks* like **Django** and **FastAPI** along with ORM Libraries like **SQLAlchemy**.
- *Implemented and invoked APIs* with **OAuth** and **mTLS** authentication.
- *Created software applications in Test-Driven Development (TDD)* environment using PyTest and UnitTest.
- *Worked on integration* and deployment tools like GitHub, Jenkins and Docker.
- *Provided support for incidents*, contribute to bug fixes, ensuring a stable and reliable application.
- *Worked on AWS monitoring* (**CloudWatch**, CloudTrail) and infrastructure as a code service (Cloud Formation and Terraform).

SKILLS

Programming Languages: Python, Java, JavaScript.
Tools & Platforms: GitHub, GitHub Actions, Jenkins, Linux, Docker, Terraform, JIRA, Confluence.
Cloud & AWS Services: EC2, Lambda, API Gateway, SQS, SNS, Glue, S3, RDS, Redshift, IAM,Kinesis, Step Functions, DynamoDB, AuroraDB, Cloud Watch, Event Bridge.
API & Web Services: Swagger, OpenAPI, Postman.
Databases: MySQL, PostgreSQL, AWS Aurora DB, DynamoDB, VectorDBs – Pinecone, Redis
AI & Machine Learning: TensorFlow, PyTorch, scikit-learn, OpenAI GPT, LLM, LangChain

EDUCATION

Trent University, **Peterborough, ON (M. Sc., Applied Modelling & Quantitative Analysis - Big Data)** **GPA: 4.0 / 4.0** **May 2022 - Sept 2023**
Coursework: *Data Science with Python, Artificial Intelligence, Data Mining, Data Visualization, Machine Learning*
Punjab Technical University, **Punjab, India (B.Tech., Computer Science)** **GPA: 8.12 / 10.0** **July 2015 - June 2019**
Coursework: *Data Structures and Algorithms, Operating Systems, Networking, Relational Database Management Systems*

ACHIEVEMENTS AND CERTIFICATIONS

- Deloitte Applause Award.
- **AWS Certified Associate Developer**.
- **ISC2 Cybersecurity Certified.**
- AWS Certified Cloud Practitioner.
- Microsoft Azure Fundamentals.
- Microsoft Azure Data Fundamentals.
- Microsoft Azure AI Fundamentals.